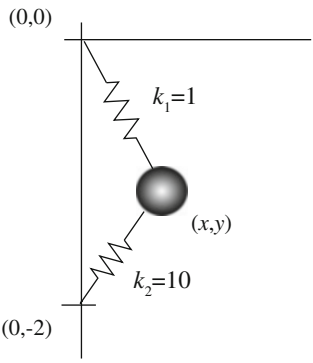


Fig. 3.12 Pendulum model with two different natural frequencies

springs are assumed to have rest lengths of ℓ_1, ℓ_2, respectively. The potential energy function is

$$U(x, y) = \frac{k_1}{2}(r_1 - \ell_1)^2 + \frac{k_2}{2}(r_2 - \ell_2)^2,$$

where $r_1 = \sqrt{x^2 + y^2}, r_2 = \sqrt{x^2 + (y + 2)^2}$.

We take $k_1 = 1$, $k_2 = 10$ (each spring has a rest length of 1) and the initial conditions $\boldsymbol{q}_0 = (0, -2.9); \boldsymbol{p}_0 = (.8718, 0)$ with energy E ≈ 2.235. The pendulum positions fill in a region which is approximately a disk with a hole cut out. A correct result is obtained using the Verlet method with a stepsize of $h = 0.04$. Yet, for exactly the same stepsize, using the energy projection method, the results are completely different.

In Fig. 3.13, the position of the point x, y is shown for three successive series of timesteps: (i) $0 \leq n < 25{,}000$, (ii) $25{,}000 \leq n < 30{,}000$, and (iii) $30{,}000 \leq n \leq 500{,}000$. In the first phase, the discrete trajectory appears to be filling in the correct region and without any evidence of nonphysical behavior. The performance is very similar to a Verlet method without projection during this period. In the second phase the system begins to move toward a limit cycle, i.e. an attractive periodic orbit. The presence of such limit cycles is impossible in a Hamiltonian system, thus it is evident that a nonphysical artefact has been introduced by the projection method. Finally, in the third phase, the trajectory is increasingly confined. In Fig. 3.14, the trajectory is shown for timesteps $n = 50{,}000$ to 500,000 and this is contrasted with the correct image of the configurational region. Because of the use of energy as a constraint, we have also given up this means of assessing the quality of the solution; the projected numerical method makes errors in the energy of relative size 10^{-14} or less. However, the observation here is that the energy conserving projection would be utterly useless at this stepsize, although, if we were to judge the quality of the method from the first 25,000 steps, we would see nothing amiss.